Aerospace Manufacturing Core
Study Guide

"AMCSG-102"

Introduction to Aerospace Manufacturing

Aircraft Construction

Airframe Construction

Airframe

An airframe is the main structural components of an airplane:

Image Credits:

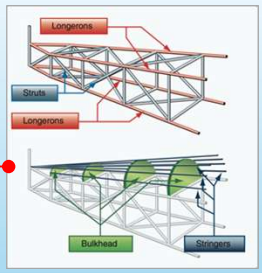
- Google "boeing.com 737 max" = (CC2) - https://commons.wikimedia.org/wiki/File:EGLF_-_Boeing_737_Max_-_N720IS_%2841646299740%29.jpg

Text from ChatGPT

Fuselage Construction: Three Types

Truss Structures

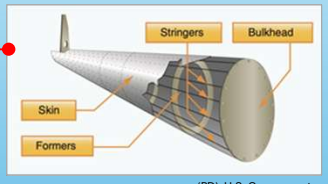
- **Material:** Wood, steel or aluminum
- **Design:** Rigid frame, lacks aerodynamic shape, **geodesic** too
- Older or smaller lightweight planes



(PD) U.S. Government

Semi-Monocoque:

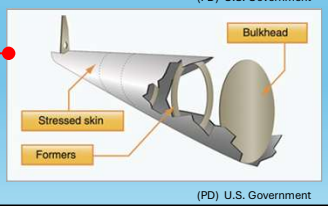
- **Material:** Aluminum or composite
- **Design:** Skin supported by frames, stringers, and bulkheads to better handle stress & damage
- Most popular today for commercial airplanes



(PD) U.S. Government

Monocoque Shell:

- **Material:** Plywood, fiberglass, composites
- **Design:** Single integrated shell is the structure (no stringers)
- Difficult to build, less common for commercial airplanes (787)



(PD) U.S. Government

Image Credits:

- Truss Image = US Gov = Public Domain = https://commons.wikimedia.org/wiki/File:Truss-type_fuselage_structure.png
- Monocoque Image = US Gov = Public Domain = https://commons.wikimedia.org/wiki/File:Monocoque_fuselage_design.png#filelinks
- Semimonocoque Image = US Gov = Public Domain = https://commons.wikimedia.org/wiki/File:Semimonocoque_fuselage_design.png

Text from ChatGPT

Fuselage Construction: **Bulkheads** & **Formers**

- **Bulkhead**
  - **Location:** Transverse, solid disc, few
  - **Strength:** High
  - **Primary Purpose:** Pressure seal, heavy loads, attach wings or tails
  - **Secondary Purpose:** Mounting longerons & stringers, shape fuselage
- **Formers** (almost same as bulkhead)
  - **Location:** Transverse, open ring, many
  - **Strength:** Medium
  - **Primary Purpose:** Mounting longerons & stringers, shape fuselage

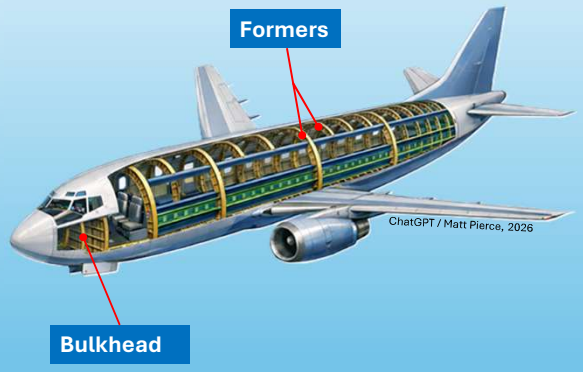


Image Credits:

- ChatGPT = licensed to me “make a diagram showing the airframe components bulkheads, formers, longerons and stringers in semi-monocoque boeing jet. Use callouts.”

Text from ChatGPT

Fuselage Construction: **Longerons** & **Stringers**

- **Longerons**
  - **Location:** Longitudinal, large, few
  - **Strength:** High
  - **Primary Purpose:** Main bending/torsion loads
- **Stringers**
  - **Location:** Longitudinal, small, many
  - **Strength:** Low
  - **Primary Purpose:** Local stiffening, skin support

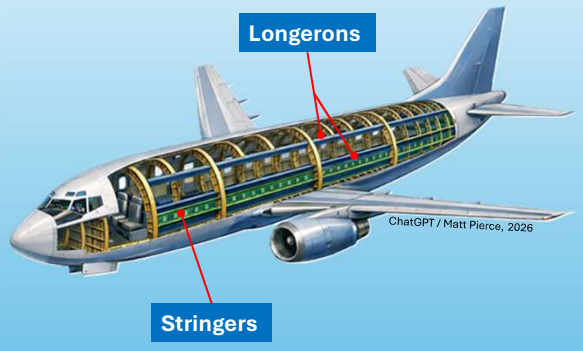


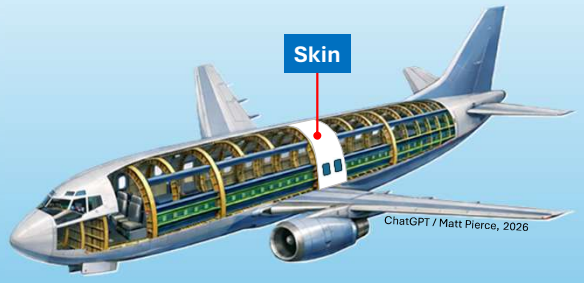
Image Credits:

- ChatGPT = licensed to me “make a diagram showing the airframe components bulkheads, formers, longerons and stringers in semi-monocoque boeing jet. Use callouts.”

Text from ChatGPT

Fuselage Construction: **Skin** & **Fasteners**

- **Skin**
  - **Location:** External shell, many small sections
  - **Strength:** Moderate
  - **Primary Purpose:** Aerodynamic shape, pressure seal, strengthens airframe
- **Fasteners**
  - **Location:** Everywhere!
  - **Strength:** Low, Moderate, Strong
  - **Primary Purpose:** Bolting together large components heavy load, screws or rivets to fasten smaller components under low to moderate load



Rivets	Bolts	Screws	Adhesives
(Ad) Aircraft Spruce	(Ad) GStatic	(Ad) Sky Geek	(Ad) Fibre Glast Devs

Image Credits:

- ChatGPT = licensed to me “make a diagram showing the airframe components bulkheads, formers, longerons and stringers in semi-monocoque boeing jet. Use callouts.”
- Rivet photo = (Ad) - Shopping at AircraftSpruce and Specialty “https://www.aircraftspruce.com/catalog/pnpages/AN470A-6-10.php?utm_source=google&utm_medium=organic&utm_campaign=shopping&utm_term=AN470A-6-10”
- Titanium Bolt Photo = (Ad) - Shopping - https://encrypted-tbn0.gstatic.com/shopping?q=tbn:ANd9GcQdkwBTBZcncE7A63si5mJcwRQQS72hApd2RD5iTzkQFr4ua48-oC8Do-vu0cnXqGRIoWqi3q--dKeiOoJQcRuvnD8AlVBMMCkEJ5LQ_6o9phYkFii6LyA3JjU
- Screw Photo = (Ad) - Shopping at SkyGeek - https://cdn11.bigcommerce.com/s-jifykode7m/images/stencil/1000w/products/101668/169815/SGP139998_1_main__23195.1698700520.jpg?c=1
- Adhesive Photo = (Ad) - Shopping at Fibre Glast Developments - https://encrypted-tbn3.gstatic.com/shopping?q=tbn:ANd9GcQ3OPpIE6lBXNh4Iw_fcBRR1Lm18

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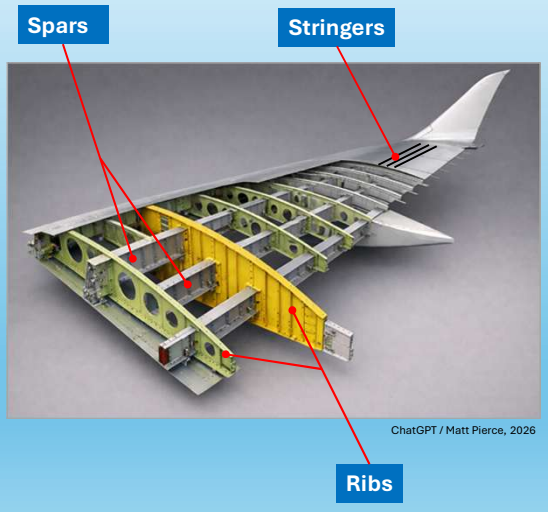
Wing Construction

Text from ChatGPT

Wing Construction: **Spars, Ribs, & Stringers**

- **Spars**
  - Parallel to wing, at least two (front/rear)
  - High strength, flexible
  - Main structural part of wing
- **Ribs**
  - Perpendicular to wing, many
  - Structural cross members (leading to trailing edge)
  - Creates shape and contour of wing
  - Transfer loads from skin to spars and stringers
- **Stringers**
  - Parallel to wing, many (ChatGPT not draw, so I mocked black line)
  - Upper and lower stringers exist
  - Attachment points for wing skin
  - Not load bearing



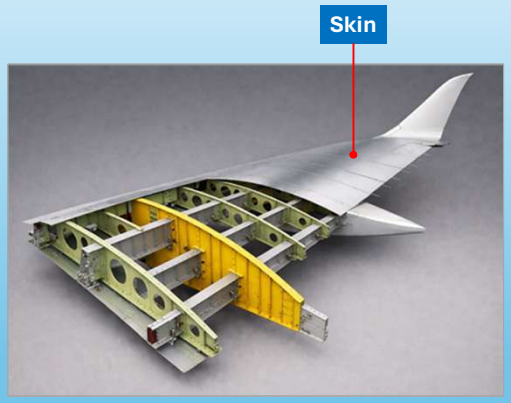
ChatGPT / Matt Pierce, 2026

- Wing Diagram: Chat GPT, many passes and dead ends…this was the best combination…no copyright for AI gen drawings per their documentation…I did this on home computer outside of work.
  - "draw an a boeing commercial airplane wing construction showing spar, longerons, ribs, and skin"
  - "remove the labels and arrows. show wing spars, wing stringers, wing ribs, and wing skin"

Text from ChatGPT

Wing Construction: **Wing Skins**

- **Wing Skin**
  - Many small sections, riveted on
  - Aerodynamic shape critical to provide lift
  - Light, rigid, easy to bend and shape
- **Types:**
  - **Aluminum** Wing Skin
    - Honeycomb layer sandwiched between upper/lower sheets
    - Less expensive than composite
  - **Carbon Fiber** Wing Skin
    - Stronger than metallic skins, lasts longer too
    - More expensive to build
  - **Fiberglass** Wing Skin
    - Smaller surfaces: ailerons, flaps



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- Wing Diagram

Factory Safety

Factory Safety (1 of 2)

- Many safety rules specific to use of tools…
- Always wear eye protection in the factory
  (flying particles, molten metal, dangerous liquids, etc.)
- May need head protection
- May need ear protection
- May need hand protection
- May need foot protection
- Work Instructions indicate what safety equipment to wear

(Ad) Aircraft Spruce & Specialty

(Ad) Mechanix Wear

(Ad) Premier Supplies

(Ad) Lehigh Safety Shoes

(Ad) PilotMall.com

- Eye protection: Shopping "Aircraft Spruce and Specialty" = https://encrypted-tbn1.gstatic.com/shopping?q=tbn:ANd9GcSRlDnva0_BDpWJ1StbgkAHFAFUIV5zJQJ1eqTTgYTJG8N2D0jLb6mkOC9Snevt92ebeGzBXdi2Xyb9Epjd8gn4UdsOdDdb

- Ear Protection "PilotMall.com" shopping site: https://encrypted-tbn1.gstatic.com/shopping?q=tbn:ANd9GcSzYgm2YBUSVHYbdtCbpswmj9BITisaFih-V03AiJndE-FATBS2EuHyORPk4eXg-XMzrs4DqQuhzlmDWRhbc-4pwFrVcvVL

- Head Protection: "Bump cap" for sale at Premier supplies website: "https://content.etilize.com/Alternate-Image1/1068706382.jpg"

- Hand Protection: Mechanix Wear website for sale. "https://encrypted-tbn1.gstatic.com/shopping?q=tbn:ANd9GcTstZGsuJ_IP29iP59mYuWnp5ko2iNCmGJ-5j5TVdPtLwdf6HuUuWrmqW0vrGYY2il0Zrb4yUQ72qtatkypeZxprfhjQZIlp3f_kk4YHhSG"

- Foot Protection (steel towed boots?): Lehigh safety shoes (Aluminum toe! Cool). “https://encrypted-tbn2.gstatic.com/shopping?q=tbn:ANd9GcR-b3Jm5JjFy_NB-h3552UifzvVH7Nj3moSIyIvU1Nd5A7utdBH9rEO-pzaNf2qf4LBTWKe6IyFjCYA0R8V2gyqVIrzBsrMf6ORBA1YSzU65NPv0G0IuxWwVw”

Factory Safety (2 of 2)

- Material Safety Data Sheet (MSDS)
  - Physical Data (melting/boiling/flash-points)
  - Toxicity, First aid, Fire-fighting, …
- Physical Hazards:
  - Fire, Explosion, Chemical Exposure, Freezing
- Health Hazards:
  - Irritants, Toxins, Carcinogens
  - Acute vs. Chronic
- Exposure Types:
  - Skin contact, Inhalation, Ingestion, Injection

(PD) U.S. Government

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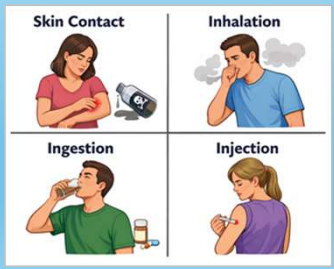

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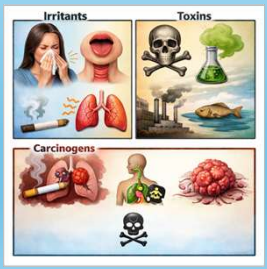

ChatGPT / Matt Pierce, 2026

Image Credits:

- MSDS Sheet Pic = Google “vector msds logo” = (CC0) - https://www.rawpixel.com/image/11071842/photo-image-fire-public-domain-red
- ChatGPT – Physical Hazards: “make a picture showing chemical exposure leading to explosion leading to fire
- ChatGPT – Health Hazards: “make a picture showing health hazards including irritants, toxins, carcinogens and what the difference is between them using pictures not words”
  + “Remove the lowest section titled differences. everything else is perfect.“
- ChatGPT – Exposure Types: “Irritants, Toxins, carcinogens”

- e

Q&A

Practice Q&A's (Q1-Q8)

Q1: What are the five components of an airframe?

A: Fuselage, wings, tail assembly, elevator, and landing gear

Q2: What type of fuselage construction were old 1917-era biplanes?

A: Truss

Q3: What is the most common type of fuselage since the 1940s (typically aluminum)?

A: Semi-Monocoque

Q4: What is the type of fuselage typically used in composite airplanes?

A: Monocoque

Q5: What transverse, solid disc assembly provides a pressure seal, attachment points for wings and longerons, and supports heavy loads?

A: Bulkhead

Q6: What open ring assembly is like a bulk-head, but has an open concentric middle?

A: Former

Q7: What large longitudinal (nose to tail) part manages bending and torsion loads?

A: Longerons

Q8: What small and plentiful part provides local stiffening and skin support?

A: Stringers

Practice Q&A's (Q9-Q16)

Q9: What part forms the external shell, providing aerodynamic shape, a pressure seal, and strengthens the airframe?

A: Skin

Q10: What part is used all over and holds everything together?

A: Fasteners

Q11: What is the main structural part of a wing that runs parallel and is high strength and flexible?

A: Spars

Q12: What is the smaller part of a wing that also runs parallel, is not load bearing, but provides skin attachment points?

A: Stringers

Q13: What is the main structural part of a wing that runs perpendicular and gives it the contour shape?

A: Ribs

Q14: What covers the wing providing aerodynamic shape for lift, is light, rigid and easily bends?

A: Skin

Q15: What are the three most common current types of wing skin?

A: Aluminum, Fiberglass, and Carbon Fiber

Q16: What type of hazard are fire, explosion, chemical exposure, and freezing?

A: Physical hazards

Practice Q&A's (Q17-Q24)

Q17: What type of hazards are irritants, toxins, and carcinogens?

A: Health hazards

Q18: What types of exposure can lead to physical and health hazards?

A: Skin contact, inhalation, ingestion, and injection

Q19: What are some basic configurations of an airplane?

A: Location of wing (high, mid, low). Number of engines. Design of tail section (high, mid, low horizontal stabilizer).

Q20: What wing type allows for shorter take-off?

A: High-wing (higher lift at low speeds)

Q21: What tail type allows for shorter take-off?

A: Low-tail (lower speed lift-off / less turbulent air)

Q22: Which wing location has a wing box connection for strength?

A: High- and low-wing. Mid-wing cannot have a box thru the "tube", so must build them integrated robustly without a wingbox.

Q23: Why are landing gear built to retract (expensive)?

A: To reduce drag

Q24: Which type of landing gear is cheaper and older, but more difficult to handle on the ground?

A: Conventional (that's why you see tricycle on all commercial aircraft…except seaplanes)

Practice Q&A's (Q25-Q32)

Q25: Where can powerplants by attached?

A: Wings, fuselage, or even the tail

Q26: What type of wing is the most stable?

A: High-wing.

Q27: What are the two primary power plant configurations?

A: Single-engine and multi-engine (2, 3, 4…)

Q28: What does PPE stand for?

A: Personal Protective Equipment (like glasses, face mask, respirator, etc.)

Q29: Federal agency for work hazard info?

A: OSHA – Occupational Safety & Health Admin

Q30: Federal agency from course 101 that oversees all Aviation in the U.S.?

A: FAA (Federal Aviation Administration)

Q31: What document identifies the PPE you should use for a given sealant application process?

A: Engineering Process Document

Q32: What should you do in case of a chemical harm incident?
A: Follow MSDS (Material Safety Data Sheet) instructions on what to do in emergency (e.g.: wash eye for 5 min., etc.)

Thank You

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https://github.com/DataResearchLabs/study_guide_intro_aerospace_mfg